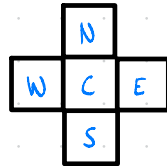


Iteratives Modell



Steady State Bedingungen...

Homogenes K, ohne Recharge: $\nabla^2 h = 0$

$$\nabla^2 h = \frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} \rightarrow \text{Krümmung in die Richtungen}$$

$$\nabla^2 h = \frac{\Delta(h)}{\Delta x^2}$$

$$\left. \begin{aligned} \rightarrow \left(\frac{\partial h}{\partial x}\right)_E &= \frac{h_E - h_C}{\Delta x} & \rightarrow \left(\frac{\partial h}{\partial y}\right)_N &= \frac{h_N - h_C}{\Delta y} \\ \rightarrow \left(\frac{\partial h}{\partial x}\right)_W &= \frac{h_C - h_W}{\Delta x} & \rightarrow \left(\frac{\partial h}{\partial y}\right)_S &= \frac{h_C - h_S}{\Delta y} \end{aligned} \right\} \text{Änderung}$$

$$\left. \begin{aligned} \frac{\partial^2 h}{\partial x^2} &\approx \left(\frac{\partial h}{\partial x}\right)_E - \left(\frac{\partial h}{\partial x}\right)_W = \frac{h_E - 2h_C + h_W}{\Delta x^2} \\ \frac{\partial^2 h}{\partial y^2} &\approx \left(\frac{\partial h}{\partial y}\right)_N - \left(\frac{\partial h}{\partial y}\right)_S = \frac{h_N - 2h_C + h_S}{\Delta y^2} \end{aligned} \right\} \text{Änderung der Änderung}$$

$$\Delta x = \Delta y = 10 \text{ m}$$

$$\nabla^2 h = \frac{h_E + h_W + h_N + h_S - 4h_C}{\Delta x^2} = 0 \rightarrow h_C = \frac{h_E + h_W + h_N + h_S}{4}$$

Homogenes K, mit Recharge: $-\nabla^2 h = R$

$$\rightarrow \nabla^2 h + \frac{R}{K} = 0$$

$$= \frac{h_E + h_W + h_N + h_S - 4h_C}{\Delta x^2} + \frac{R}{K} = 0$$

$$\rightarrow h_C = \frac{h_E + h_W + h_N + h_S}{4} + \frac{\Delta x^2}{4} \frac{R}{K}$$

Inhomogenes K, ohne Recharge: $\nabla \cdot (K \nabla h) = 0$

$$\left. \begin{aligned} \rightarrow K_E \left(\frac{\partial h}{\partial x}\right)_E &= K_E \frac{h_E - h_C}{\Delta x} \\ \rightarrow K_W \left(\frac{\partial h}{\partial x}\right)_W &= K_W \frac{h_C - h_W}{\Delta x} \\ \rightarrow K_N \left(\frac{\partial h}{\partial y}\right)_N &= K_N \frac{h_N - h_C}{\Delta y} \\ \rightarrow K_S \left(\frac{\partial h}{\partial y}\right)_S &= K_S \frac{h_C - h_S}{\Delta y} \end{aligned} \right\} \nabla \cdot (K \nabla h) = \frac{K_N(h_N - h_C) + K_S(h_S - h_C) + K_E(h_E - h_C) + K_W(h_W - h_C)}{\Delta x^2} = 0$$

$$= K_N h_N + K_S h_S + K_E h_E + K_W h_W - h_C(K_N + K_S + K_E + K_W) = 0$$

$$\rightarrow h_C = \frac{K_N h_N + K_S h_S + K_E h_E + K_W h_W}{K_N + K_S + K_E + K_W} = \frac{\text{numerator}}{\text{denominator}}$$

(Homogener fall: $h_C = \frac{h_N + h_S + h_E + h_W}{4}$)

Inhomogenes K, mit Recharge: $-\nabla \cdot (K \nabla h) = R$

$$\rightarrow \nabla \cdot (K \nabla h) + R = 0$$

$$K_N(h_N - h_C) + K_S(h_S - h_C) + K_E(h_E - h_C) + K_W(h_W - h_C) + R \Delta x^2 = 0$$

$$= K_N h_N + K_S h_S + K_E h_E + K_W h_W - h_C(K_N + K_S + K_E + K_W) + R \Delta x^2 = 0$$

$$\rightarrow h_C = \frac{K_N h_N + K_S h_S + K_E h_E + K_W h_W + R \Delta x^2}{K_N + K_S + K_E + K_W} = \frac{\text{numerator} + R \Delta x^2}{\text{denominator}} = \frac{\left[\frac{\text{m}^2}{\text{day}}\right] + \left[\frac{1}{\text{day}}\right] \cdot \left[\text{m}^2\right]}{\left[\frac{\text{m}}{\text{day}}\right]} = [\text{m}]$$

Darcy's Law

$$q = -K \cdot \nabla h = -K \cdot \frac{\partial h}{\partial l}$$

$$R = \frac{R_{in}}{1000 \cdot b}$$

q: Darcy Geschwindigkeit $\left[\frac{\text{m}}{\text{day}}\right]$

K: hydraulische Leitfähigkeit $\left[\frac{\text{m}}{\text{day}}\right]$

R: Modell-Recharge rate $\left[\frac{1}{\text{day}}\right]$

R_{in}: Input-Recharge rate $\left[\frac{\text{mm}}{\text{day}}\right]$

b: aquifer thickness [m]